

What DDI people would ask BlueCat LiveAssist

10 personas mapped to their LiveAssist DDI use cases. Each persona question is paired with the decision it feeds, how often it comes up, and the functional topic(s) it falls under. **Click any ask** to open the full scenario — the situation that produced it, the pressure behind it, and what came back through LiveAssist via the Integrity MCP server. ⚡ marks the frontier — reads that feel like writes (simulation, blast radius, what-if).



DNS

DHCP

IPAM

Security

Automation & Cloud

Platform Ops

Access & Governance

M&A / Integration

Agent & AI

TOPIC MIX BY PERSONA — CLICK A ROW TO FILTER

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P1—NOC / Tier-1 service desk · Highest read volume in the building, least equipped to use the BAM UI.

#1

"What is 10.42.7.19?"

First move on almost every network ticket; identity before anything else.

many/day

IPAM

DNS

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THE SITUATION

A P2 ticket lands with nothing but an IP in the subject line: "10.42.7.19 is doing something weird." The Tier-1 analyst has no hostname, no owner, no location, and no idea whether this is a printer, a domain controller, or someone's Raspberry Pi.

WHY IT MATTERS

Every downstream decision — severity, routing, whether to page anyone — depends on knowing what the thing is. Today that means the BAM tree view, a subnet the analyst has to guess at, and usually a Slack message to whoever answers first. The ticket sits in triage for twenty minutes before anyone knows if it matters.

WHAT LIVEASSIST RETURNED

The analyst asks LiveAssist in plain language. Through the Integrity MCP server it returns the address object with its host record, MAC, current DHCP lease and lease history, the parent subnet and block, the assigned owner and location fields, and every DNS record that resolves to it. It is a domain controller in the Frankfurt datacenter subnet. The ticket is re-severitied and routed to the AD team in under a minute, with the identity pasted into the notes as evidence.

#2

"Is this record still real, or a ghost?"

Decides escalate vs. close-as-stale — the most common wrong call at Tier-1.

daily

DNS

IPAM

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THE SITUATION

A monitoring alert flags a name that has not answered in days. The analyst has to decide: is this a real service that broke, or a leftover record for something decommissioned last quarter that nobody cleaned up?

WHY IT MATTERS

Closing a live outage as stale is the worst call Tier-1 makes, and escalating a ghost record burns an engineer's afternoon. The signal that separates the two — is anything actually using this address — is scattered across DHCP leases, deployment state, and change history that Tier-1 has no access to.

WHAT LIVEASSIST RETURNED

LiveAssist answers with the record's state in one pass, read live from Integrity: whether the address has an active lease, when the lease last renewed, when the record was created and last modified, whether it is deployed to the serving BDDS or only staged in BAM, and whether the reverse PTR still exists. No lease in four months, created by a decommissioned automation account, PTR already gone. The analyst closes it as stale and files it to the hygiene queue with the evidence attached.

#3 "What does this name resolve to, in every view?"

Split-horizon mismatch is the #1 source of "works for me" false alarms.

many/day

DNS

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THE SITUATION

A user reports that an internal application URL works from the office and fails from the VPN. The service owner insists nothing changed. Tier-1 is caught between two people who are both correct from where they are standing.

WHY IT MATTERS

Split-horizon DNS means the same name legitimately has different answers in different views, and the failure mode is that someone edited one view and forgot the other. Confirming this from the UI means opening each view separately and comparing by eye, which nobody at Tier-1 does reliably under ticket pressure.

WHAT LIVEASSIST RETURNED

One question to LiveAssist returns the name resolved across every configured view at once, side by side, with the record type and TTL for each. The internal view points at the new load balancer VIP; the VPN-facing view still points at the decommissioned host. The mismatch is visible in the answer itself, and Tier-1 escalates with a specific, provable finding rather than a symptom report.

#4 "Who owns this subnet — team, contact, change window?"

Routes the ticket correctly without a round of Slack archaeology.

many/day

IPAM

Access & Governance

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THE SITUATION

A change is needed on a subnet the analyst has never seen before. Who owns it? Which team's change window applies? Is there a standing freeze? The subnet description field says

"MIGRATED 2019 - DO NOT USE" and the analyst has no idea whether that is still true.

WHY IT MATTERS

Misrouted tickets are the largest single source of delay in the queue. The ownership information exists — in UDF fields, in tag hierarchies, in a spreadsheet somebody maintains — but reassembling it costs a round of Slack archaeology every time, and the answer is only as good as whoever replies first.

WHAT LIVEASSIST RETURNED

LiveAssist returns the subnet with its full tag path, every user-defined field including owner team and escalation contact, the parent block's inherited ownership where the subnet itself is blank, and the change-window UDF. LiveAssist resolves the inheritance so the analyst does not have to walk the tree. The ticket goes to the right team on the first hop, and the stale description is flagged for cleanup.

#5

"Show me the last 5 changes touching this IP or name."

Answers "did we break it?" before escalating to an engineer.

daily

DNS

IPAM

Platform Ops

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THE SITUATION

A service degraded twenty minutes ago. Before anyone wakes up an engineer, the question in the room is the obvious one: did we do this to ourselves?

WHY IT MATTERS

The change log exists in BAM, but reading it means knowing which object to look at, and the object in question may be the IP, the host record, the parent subnet, or the zone. Tier-1 typically cannot reconstruct that chain, so the default is to escalate and let an engineer check — which is a fifteen-minute round trip for a question that should take seconds.

WHAT LIVEASSIST RETURNED

The analyst asks for recent changes touching the IP or the name and LiveAssist walks the related objects for them, returning the last several modifications across the address, its host records, and the enclosing subnet — each with timestamp, the account that made it, and what field changed. A TTL was dropped to 30 seconds by an automation account eighteen minutes before the alert. That correlation goes into the ticket and the escalation arrives at the engineer already half-solved.

#6

"Their laptop can't get an address — what happened to this MAC in the last hour?"

Lease-level forensics without needing DHCP server log access.

daily

DHCP

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THE SITUATION

A user at a branch office cannot get on the network. Their laptop shows a 169.254 self-assigned address. The desk-side technician wants to know whether the DHCP server ever saw this machine at all.

WHY IT MATTERS

Answering this properly means DHCP server log access, which Tier-1 does not have and should not have. Without it the ticket becomes a guessing game between "bad port," "exhausted scope," and "MAC filtered," and the usual resolution is to dispatch someone to swap a cable.

WHAT LIVEASSIST RETURNED

LiveAssist returns the lease history for that MAC across the estate: every DISCOVER and OFFER it can account for, which scope responded, what address was offered, and when the last successful lease expired. The MAC pulled an address on the guest VLAN forty minutes ago and is now presenting on the corporate VLAN where it has no reservation. The problem is a mis-patched port, identified without dispatching anyone and without giving Tier-1 log access.

#7

"Give me a plain-English summary of everything BAM knows about this host."

One paste into the ticket. The single highest-volume LiveAssist-native ask on this page.

many/day

DNS

DHCP

IPAM

Agent & AI

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THE SITUATION

Every ticket the analyst opens starts the same way: fifteen minutes of assembling a picture of one host out of six different screens, then retyping the useful parts into the ticket so the next person does not have to repeat it.

WHY IT MATTERS

This is the highest-volume work in the building and none of it is analysis — it is transcription. The BAM UI can show every piece of it, but never on one screen and never in a form that can be pasted into a ticket. The cost is not any single lookup; it is the same lookup performed several hundred times a day across the desk.

WHAT LIVEASSIST RETURNED

One question to LiveAssist produces a written summary: what the host is, where it lives in the address hierarchy, who owns it, its DNS records across all views, its DHCP lease and reservation state, when it last changed and by whom, and anything anomalous — a missing PTR, a lease that has not renewed, a name that resolves differently internally and externally. The analyst reads it, pastes it into the ticket, and starts on the actual problem. This is the ask with no UI equivalent at any price, because the deliverable is prose, not a record.

P2 — DNS operations engineer · Highest-frequency hands-on role; read-only serves pre-flight and post-flight.

#8

"Does this name exist anywhere — any zone, any view — before I promise it?"

Pre-flight on every request; prevents the collision you find out about at deploy.

many/day

DNS

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THE SITUATION

An application team requests `app-prod-07.corp.example.com` for a new deployment. The DNS engineer needs to promise that name is available before the team starts building against it.

WHY IT MATTERS

"Available" means available in every zone and every view, including the ones this engineer does not normally touch and the ones another team owns. Checking that properly means several searches, and the failure mode — discovering the collision at deploy time, after the app team has hardcoded the name — is expensive and public.

WHAT LIVEASSIST RETURNED

A single LiveAssist existence check across the whole namespace returns every match: zone, view, record type, current target, and whether it is deployed or staged. The name is free in the internal views but exists as a CNAME in the DMZ view pointing at a partner-facing endpoint that nobody remembers creating. The engineer denies the name, proposes an alternative, and opens a separate ticket on the mystery CNAME — all before the app team writes a line of config.

#9

"Everything that points at 10.1.2.3 — A, PTR, CNAME chains, MX/SRV targets."

Impact check before touching an address. Needs traversal, not a flat filter.

daily

DNS

Hide the story ▲

THE SITUATION

An address is being reclaimed for a re-IP. Before it moves, the engineer needs the complete set of things that reference it — not just the obvious A record.

WHY IT MATTERS

DNS references form a graph. An address has an A record; that name has CNAME aliases; it may be an MX target, an SRV target, or a nameserver record for a delegated zone. A flat search for the IP finds the A and the PTR and misses everything that points at the name rather than the address. That gap is how a mail flow breaks during a routine renumber.

WHAT LIVEASSIST RETURNED

LiveAssist traverses the reference graph rather than filtering a table: the A and PTR, every CNAME chain terminating at that name, every MX and SRV record whose target resolves there, and any NS delegation. Three CNAMEs and an SRV record for the internal SIP service turn up that no flat search would have found. The re-IP plan grows by four line items and does not take down voice.

#10

"What breaks if I delete this record?"

The question that prevents outages. Nobody can answer it reliably today.

daily

frontier

DNS

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THE SITUATION

A cleanup ticket asks the engineer to delete an old record. It looks unused. It has looked unused for a year. The engineer has been burned before by exactly this record shape.

WHY IT MATTERS

Nobody can answer "what breaks" reliably, so the honest answers are either "delete it and find out" or "leave it forever." Most estates choose the second, which is why hygiene backlogs grow without bound. The information needed — inbound references, dependent chains, deployment scope — exists but has never been assembled into an impact answer.

WHAT LIVEASSIST RETURNED

The engineer asks what breaks. LiveAssist returns a dependency-ordered impact set: every record that references this one directly, everything that references those, which zones and which serving BDDS instances are affected, and which of the dependents are themselves live versus already stale. The record has one live dependent — a CNAME used by a legacy backup job. The deletion is deferred, the backup job owner is notified, and the outage that would have followed simply does not happen.

#11 "Hygiene sweep: missing PTRs, dangling CNAMEs, duplicate name/type pairs."

Generates the cleanup queue instead of waiting for a complaint.

daily/weekly

DNS

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THE SITUATION

Rather than waiting for the hygiene backlog to announce itself through a complaint, the engineer wants the cleanup queue generated for them at the start of the week.

WHY IT MATTERS

Missing PTRs break reverse lookups in logs and security tooling. Dangling CNAMEs are takeover surface. Duplicate name/type pairs produce nondeterministic answers. Every one of these is findable by query, and none of them is findable by looking at the UI, so in practice they are found by whoever gets paged.

WHAT LIVEASSIST RETURNED

A single LiveAssist sweep returns the three classes at once, scoped by zone and ranked by blast radius: forward records with no matching PTR, CNAMEs whose targets no longer resolve, and name/type collisions. The output is a work queue with object references, not a report to be re-derived. The engineer triages it in twenty minutes instead of reacting to it over the following six weeks.

#12 "Does BAM's intent match what the servers actually answer?"

Drift between IPAM and the wire is invisible until it bites.

daily

DNS

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THE SITUATION

An intermittent resolution failure has been open for a week. BAM says the record is correct. The engineer suspects BAM and the servers disagree, but proving it means dig-ing every serving instance by hand.

WHY IT MATTERS

BAM holds intent. The BDDS instances hold what is actually served. Drift between the two is invisible from either side alone, and it is produced by ordinary events — a failed deploy, a manual edit on the appliance, a partial replication. The gap can persist for months because nothing routinely compares them.

WHAT LIVEASSIST RETURNED

LiveAssist resolves the name against each serving instance and diffs the live answers against BAM's intent, per view, per server. Two of five BDDS instances are still serving the previous A record because the last deploy partially failed and the failure was logged but never surfaced. The engineer redeploys the two, and the intermittent failure — which depended on which resolver the client happened to hit — disappears.

#13

"What DNS changes are staged but not deployed, and to which servers?"

Decides deploy-now vs. hold; also catches someone else's half-finished work.

daily

DNS

Hide the story ▲

THE SITUATION

It is 4pm on a Thursday and the engineer is deciding whether to run a deploy before the weekend. The question is not whether their own changes are ready; it is whether someone else has left something half-finished in the queue.

WHY IT MATTERS

A deploy publishes everything staged, not just your work. Somebody else's abandoned mid-edit goes out with it. The staged set is visible in BAM but not in a form that answers "whose is this and is it safe," so the practical policy at most sites is to deploy and hope, or to freeze and fall behind.

WHAT LIVEASSIST RETURNED

The engineer asks what is staged but not deployed, and LiveAssist returns the pending change set grouped by object and by author, with target servers and the age of each change. Fourteen changes are the engineer's own; three belong to a colleague who is on PTO and left a zone mid-restructure. The deploy is scoped to exclude that zone, the colleague's work stays staged, and nothing goes out that nobody can explain.

#14 "Which zones failed to deploy — and why?"

The morning queue. Status is easy; the reason is what's actually needed.

daily

DNS

Hide the story ▲

THE SITUATION

The engineer's morning starts with the deploy queue. Three zones show failed. The status is easy to see; the reason is not, and without it the only recourse is to retry and watch.

WHY IT MATTERS

Deploy failures have distinct causes — a syntax error in a record, an unreachable serving instance, a signature problem, a permissions issue — and they need entirely different responses. Retrying a permissions failure wastes the morning. The reason exists in the deployment logs, one appliance at a time.

WHAT LIVEASSIST RETURNED

LiveAssist pulls the deployment detail out of Integrity and returns the failed zones with the actual failure reason attached to each: one zone has a record with an invalid target, one BDDS was unreachable during the window, and one failed DNSSEC signing because a key had rolled. Three different fixes, identified before the first coffee, instead of three retries and an afternoon of narrowing it down.

#15 "What changed in DNS in the 30 minutes around this outage?"

Correlates the change log to an incident timeline. Needs a time window, not a list.

incident

DNS

Hide the story ▲

THE SITUATION

A major incident is being reconstructed. The service failed at 14:12. The incident channel wants to know what changed in DNS anywhere near that moment — not what changed today, and not what changed to this specific object, because nobody yet knows which object matters.

WHY IT MATTERS

Change logs answer "what changed to this thing." Incident reconstruction needs the inverse: "what changed at this time, across everything." Assembling that today means exporting change history and filtering it in a spreadsheet while the bridge waits, which is why the change correlation usually arrives after the incident is resolved and the lesson is lost.

WHAT LIVEASSIST RETURNED

The engineer gives LiveAssist a time window and a scope. It returns every DNS change in the estate between 13:42 and 14:42, ordered by timestamp, with object, author, and before/after values. A TTL reduction and a nameserver change landed six minutes apart, eight minutes before the failure. The incident timeline gets its first real anchor while the bridge is still live.

#16

"Show me records nobody has queried in 12 months."

The retirement candidate list. Requires query telemetry BAM doesn't hold.

quarterly

DNS

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The zone has four thousand records. The engineer is confident that most of them are dead, and has no way to prove which ones, so all four thousand stay.

WHY IT MATTERS

"Nobody has asked for this name in a year" is the only trustworthy retirement signal, and it lives in DNS query telemetry — not in BAM. Without it, retirement decisions are made on record age and gut feel, both of which are wrong often enough that the safe policy is to never delete anything.

WHAT LIVEASSIST RETURNED

With query telemetry joined to the record inventory, LiveAssist returns records with no observed queries in the trailing twelve months, annotated with creation date, owner, and whether anything else in DNS still references them. Eleven hundred records qualify and four hundred of those have zero inbound references. That is a defensible retirement list, produced by a query instead of a project.

P3 — DHCP / access-network engineer · Bursty; hardest questions are forensic and temporal.

#17

"How full is this scope, and which scopes estate-wide are about to run dry?"

Capacity triage before the pager, not after.

daily

DHCP

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Friday afternoon, and the engineer wants to know which DHCP scopes are going to page someone over the weekend before anyone leaves for the day.

WHY IT MATTERS

Scope exhaustion is entirely predictable and almost always discovered by the users who cannot get an address. Per-scope utilization is visible in the UI one scope at a time, which does not scale to an estate with hundreds of them, so nobody checks proactively and everyone reacts.

WHAT LIVEASSIST RETURNED

One question to LiveAssist returns utilization for every scope in the estate, ranked, with free-address counts and the trend over the last several weeks. Four scopes are above 90% and two of those are climbing fast — a branch office that just onboarded contractors and a lab that has been quietly leaking reservations. The engineer widens one range and cleans up the other on Friday, and nobody gets paged.

#18 "Who was on 10.20.5.66 at 14:20 yesterday?"

Attribution for abuse, incident, and legal requests. Purely a temporal read.

weekly

DHCP

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THE SITUATION

Security forwards an abuse complaint: traffic from 10.20.5.66 at 14:20 yesterday. Who was that? The address has been reassigned twice since.

WHY IT MATTERS

Attribution is the entire point of the request, and current-state IPAM cannot answer it — the address now belongs to somebody else. This is a temporal question and it comes from abuse handling, incident response, and legal discovery alike, which means the answer has to be defensible, not approximate.

WHAT LIVEASSIST RETURNED

LiveAssist answers the address at that timestamp rather than now: the lease active at 14:20, its MAC, the hostname the client presented, the scope that issued it, and the lease start and end times bracketing the moment. It resolves to a specific laptop in the Austin office. The security team gets an attribution with a lease record behind it, not a best guess from a spreadsheet.

#19 "Are there static assignments sitting inside a DHCP range?"

The classic duplicate-address outage, findable in advance.

weekly

DHCP

IPAM

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THE SITUATION

A duplicate-address outage last quarter cost the site four hours. It was caused by a statically configured server sitting inside a DHCP range. The engineer wants to know where else that is true, before it happens again.

WHY IT MATTERS

This is the most preventable outage in DHCP and the hardest to see, because the static assignment and the dynamic range live in different parts of the data model and look completely reasonable in isolation. Finding the overlaps means comparing two lists across hundreds of scopes.

WHAT LIVEASSIST RETURNED

LiveAssist cross-references static host records and reservations against every configured dynamic range and returns the collisions with the object on each side. Nine addresses across four scopes are statically assigned inside an active range, two of them on ranges with fewer than ten free addresses left — meaning the collision was weeks, not months, away. All nine are corrected in one change window.

#20 "What options does this scope inherit, and from where?"

Debugging wrong-gateway/wrong-DNS complaints; the chain matters more than the values.

weekly

DHCP

[Hide the story ▲](#)

THE SITUATION

Clients on one subnet are getting the wrong default gateway. The scope's own configuration looks correct. The engineer knows an option is being inherited from somewhere but not from where.

WHY IT MATTERS

DHCP options inherit down a hierarchy — from server to network to scope to reservation — and the effective value is the result of that chain, not any single level. The UI shows what is set at the level you are looking at, which is exactly the information that does not help. Debugging by elimination means clicking up the tree and guessing.

WHAT LIVEASSIST RETURNED

LiveAssist returns the effective option set for the scope with provenance on every value: which level each option was defined at, which levels were overridden, and where the winning value came from. The router option is being set two levels up at the network container, from a migration in 2021 that nobody unwound. The chain is the answer, and it takes one question instead of an afternoon.

#21 "Which scopes are failing to hand out addresses — is it exhaustion or config?"

Separates a capacity problem from a config problem before anyone is dispatched.

weekly

DHCP

[Hide the story ▲](#)

THE SITUATION

Complaints from one site say devices are not getting addresses. The scope shows free space. Something else is wrong, and the engineer needs to know what before dispatching a technician four hours away.

WHY IT MATTERS

"No address" has at least three unrelated causes — pool exhaustion, a failover peer out of sync, or an option or class configuration rejecting the clients — and they need different people. Dispatching for a config problem is expensive; treating a capacity problem as a config problem is worse.

WHAT LIVEASSIST RETURNED

LiveAssist returns the scope's failure profile: current utilization, free versus offered-but-not-acked counts, failover peer state and lease-database sync status, class and filter configuration, and the recent pattern of DISCOVERs that received no OFFER. The pool has space, but the failover peer has been in a partner-down state for six days and the scope is only serving its own half of the range. No dispatch, one appliance fix.

#22 **"Next free /26 that's in-region, RFC1918, and not adjacent to X."**

The daily allocation decision. Needs constraints expressed as intent, not filters.

daily

IPAM

[Hide the story ▲](#)

THE SITUATION

A new site needs a /26. It has to be in-region for the regional routing policy, RFC1918, and deliberately not adjacent to the block used by the partner VPN so a future summarization does not overlap.

WHY IT MATTERS

This is the daily allocation decision, and every constraint on it is policy rather than data structure. The current process is to list candidate free space, then manually check each candidate against the rules, then discover on the fourth candidate that it violates one. The rules live in a wiki page and in the architect's head.

WHAT LIVEASSIST RETURNED

The architect expresses the constraints as the question rather than as post-filtering. LiveAssist returns candidate blocks that already satisfy all of them, each annotated with why it qualifies — region tag, parent block, address family, and adjacency to reserved ranges. Two candidates come back, both usable. The allocation decision takes a minute and is defensible in the design review because the constraints are recorded with the answer.

#23 "Next free IP in this subnet."

The most-repeated read in IPAM, full stop.

many/day

IPAM

Hide the story ▲

THE SITUATION

A server build needs an address in a specific subnet. This is the single most repeated question in IPAM and it happens dozens of times a day across the team.

WHY IT MATTERS

The UI can do this, but it costs a login, a tree navigation, and a scroll, and it is being done constantly by people who are in the middle of something else. At volume, the friction is the whole cost — and the second-order cost is people reserving addresses out of a spreadsheet instead, which is how IPAM drifts from reality.

WHAT LIVEASSIST RETURNED

LiveAssist returns the next free address — or the next several — in the requested subnet, skipping reserved and excluded ranges and anything with a stale-but-present lease, with the current utilization included so the requester knows whether they are taking the last one. It answers in the flow of whatever the engineer was already doing, which is why the address actually gets recorded in IPAM instead of on a sticky note.

#24 "Which subnets exceed 85%, and when do they exhaust at current burn?"

Triggers procurement and re-IP projects while there's still time to plan.

daily/weekly

IPAM

Hide the story ▲

THE SITUATION

Budget planning is in six weeks and the architect needs to know whether the estate needs more address space this cycle or can wait another year.

WHY IT MATTERS

"Which subnets are over 85%" is a snapshot and does not answer the question — a subnet at 86% that has not moved in two years is fine, and one at 70% that has climbed twenty points since March is not. The trend is what matters and nothing computes it, so the answer is usually an engineer's instinct dressed up in a slide.

WHAT LIVEASSIST RETURNED

LiveAssist reads Integrity's utilization history and returns subnets above the threshold with their consumption trend over the trailing period and a projected exhaustion date at the current burn rate. Six subnets are over 85%; two of those exhaust within the budget cycle, and one of them serves the site that is doubling headcount. The procurement ask is specific, timed, and has a chart behind it instead of a hunch.

#25 "Do we have overlapping or duplicated address space?"

Blocks merges and VPNs; cheaper to find now than during a cutover.

weekly

IPAM

Hide the story ▲

THE SITUATION

A partner VPN is being stood up and the architect has learned, three times now, that overlapping address space is discovered during the cutover rather than during the design.

WHY IT MATTERS

Overlaps accumulate quietly — from acquisitions, from lab environments built off-book, from sites that were designed independently before there was a standard. They are invisible until two of them need to route to each other. By then the cost is a re-IP under time pressure.

WHAT LIVEASSIST RETURNED

LiveAssist scans the full address hierarchy for overlapping and duplicated space and returns the collisions with both sides identified, their tags, and whether each is actually in use. Two overlaps surface: one is a decommissioned lab that can simply be deleted, and one is a live site that will need renumbering. Discovering that during design costs a planning cycle; discovering it during cutover costs a weekend and a postmortem.

#26

"What can we reclaim — no lease, no DNS, no ARP, no ticket?"

Recovers space without buying any. The four-way join is what makes it trustworthy.

monthly

IPAM

DNS

DHCP

Hide the story ▲

THE SITUATION

The architect has been told there is no budget for more address space this year. The only remaining option is to find what is already allocated and not actually being used.

WHY IT MATTERS

Reclamation is only safe if the evidence is complete. An address with no DHCP lease might still be a statically configured server. One with no DNS record might still be routed and live. Any single signal produces false positives, and one bad reclamation makes the whole program politically untouchable.

WHAT LIVEASSIST RETURNED

LiveAssist joins four independent sources: no active or recent DHCP lease, no forward or reverse DNS, no recent ARP or neighbor entry from the network layer, and no open ticket or CMDB record referencing the address. Only addresses failing all four are returned. Twelve hundred addresses qualify across the estate — roughly a /22 recovered without a purchase order, and the four-way join is what makes the change board approve it.

#27

"Explain our addressing scheme — how is this estate actually organized?"

Onboarding, and the question no incumbent can answer from a tree view.

onboarding

IPAM

Hide the story ▲

THE SITUATION

A new architect joins and asks the question nobody on the team can answer cleanly: how is this estate actually organized? What does the tag hierarchy mean? Why are the EMEA sites structured differently from the Americas?

WHY IT MATTERS

The scheme is real, it is consistent within eras, and it is entirely undocumented — it lives in the shape of the data and in the memory of people who have left. The tree view shows the structure without explaining it, so onboarding takes months and every new allocation risks violating a convention nobody has written down.

WHAT LIVEASSIST RETURNED

LiveAssist reads the whole hierarchy and describes it: how blocks are subdivided, what the tag taxonomy encodes, which conventions hold estate-wide and which are regional, where the exceptions cluster, and which parts of the tree appear to follow a different generation of the standard. The output is the documentation that should have existed, generated from the data rather than from anyone's recollection — and reviewing it surfaces two conventions the incumbents had stopped noticing.

#28 "How much of the estate is IPv6-reachable, by site and by business unit?"

Adoption tracking that leadership asks for and nobody can produce.

quarterly

IPAM

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THE SITUATION

Leadership asks for the IPv6 adoption number ahead of a board update. The honest answer from the team is that nobody knows.

WHY IT MATTERS

IPv6 adoption is tracked as a program milestone but measured as a guess, because the data needed to compute it — which subnets have v6 counterparts, which have live v6 hosts, broken down by site and business unit — has to be assembled by hand from three different views and re-derived every quarter.

WHAT LIVEASSIST RETURNED

LiveAssist returns v6 coverage against the v4 estate, dimensioned by site and by business-unit tag: which networks have dual-stack allocations, which of those have actual v6 host records and leases rather than just allocations, and where the gaps concentrate. Adoption is 34% by allocation but 11% by active use, and the gap is almost entirely in two business units. That distinction is what turns a status number into a plan.

P5 — DDI platform / infrastructure ops · Owns the BAM/BDDS plumbing; likely runs this from cron.

#29 "Is everything OK right now?"

Start-of-shift glance across servers, HA, and services.

daily

Platform Ops

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THE SITUATION

Start of shift. Before picking up anything else, the platform engineer wants a single glance across the DDI estate to know whether they are walking into a normal day.

WHY IT MATTERS

The information is spread across appliance health, HA and failover pairing state, service status per instance, replication, and deploy queues — five screens, checked in a different order by every engineer, and skipped entirely on a busy morning. Problems that were visible at 8am get discovered at 11am by a user.

WHAT LIVEASSIST RETURNED

One question to LiveAssist returns the Integrity estate posture: every BDDS and BAM instance with service state and last check-in, HA pair status, DNS and DHCP service health per node, replication lag, and anything currently degraded — with the abnormal items surfaced first rather than buried in a list of green. Two minutes, one answer, and the engineer knows whether today is a maintenance day or a firefighting day.

#30 "Tell me if anything needs attention. (one cron-able pass)"

The proactive alerting BAM doesn't do for itself.

continuous

Platform Ops

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THE SITUATION

The engineer wants the inverse of the morning glance: not a dashboard someone has to remember to look at, but a scheduled pass that says nothing at all unless something needs a human.

WHY IT MATTERS

BAM does not proactively tell you when it is unhappy. The gap between a condition becoming true and someone noticing it is however long it takes for the next person to check, which on a weekend is measured in days. Existing monitoring covers reachability but not DDI-specific conditions like a stalled deploy queue or a scope about to exhaust.

WHAT LIVEASSIST RETURNED

The same read runs on a schedule through LiveAssist and produces output only when something crosses a threshold: a failed deploy, a scope over 90%, a stale mirror, an HA pair that has been degraded for more than an hour, a DNSSEC signature approaching expiry. The engineer gets a quiet inbox on normal days and a specific, actionable message on the days that matter. The proactive alerting the platform does not do for itself, assembled from reads.

#31 "Are backups current and is replication healthy?"

Backup drift is invisible until the restore you can't do.

daily/weekly

Platform Ops

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THE SITUATION

The quarterly DR review asks a simple question: if BAM were lost right now, what would be restored, and from when? Nobody wants to guess.

WHY IT MATTERS

Backup and replication drift silently. A backup job that has been failing for three weeks looks identical from the outside to one that ran last night, until the restore. Replication that has fallen behind is invisible until a failover promotes a stale secondary. Both are checkable and neither is routinely checked.

WHAT LIVEASSIST RETURNED

LiveAssist reports backup currency and replication health together: last successful backup per instance with age and size, backup target reachability, replication state and lag between primary and secondary, and the delta between what the primary holds and what the secondary would serve if promoted. One secondary is eleven days behind because a schema change broke its sync. Found in a review instead of in a failover.

#32 "What changed in the estate since yesterday?"

Drift detection and unauthorized-change detection in one read.

daily

Platform Ops

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THE SITUATION

The engineer wants to know what moved in the estate overnight — not to any particular object, but anywhere.

WHY IT MATTERS

Two different needs collapse into this one read: legitimate drift detection, so that today's baseline is understood, and unauthorized-change detection, so that something done outside process is noticed the next day rather than the next audit. Neither is served by per-object change history, which requires knowing which object to ask about.

WHAT LIVEASSIST RETURNED

LiveAssist diffs the estate against yesterday and returns everything that changed — objects created, modified, and deleted, grouped by type and by the account that made the change, with the before and after values. Most of it is the automation account doing exactly what it should. Three changes came from a named user account at 23:40, outside any change window, which is a conversation worth having the same morning rather than at the next audit.

#33

"Show me the estate exactly as it was last Tuesday."

Root-cause scoping and rollback planning. Needs real time travel, not a diff.

incident

Platform Ops

[Hide the story ▲](#)**THE SITUATION**

An incident is being root-caused two days later. The estate has changed since. The engineer does not want a change log; they want to see the configuration exactly as it stood at the moment the failure began.

WHY IT MATTERS

Change logs tell you what moved. Root cause frequently depends on what was true — the effective option set, the full record state, the scope configuration as it existed then. Reconstructing that by replaying a change log backwards is error-prone and slow, and it is exactly the work nobody has time for during a postmortem.

WHAT LIVEASSIST RETURNED

LiveAssist answers the estate as of a timestamp: the objects, their field values, and their relationships as they stood last Tuesday at 14:00. The engineer sees that the zone had a different SOA and one fewer nameserver at the moment of failure — a state that was corrected an hour later by someone who did not realize it was the cause. The postmortem gets a real finding, and the rollback plan for the next occurrence is written against a known-good snapshot.

#34

"Which objects haven't been touched in two years and have no dependents?"

Tech-debt cleanup with the safety check built into the question.

quarterly

Platform Ops

[Hide the story ▲](#)**THE SITUATION**

Tech-debt cleanup is approved but the team has no safe starting list. Old objects are easy to find; old objects that nothing depends on are the ones that can actually be deleted.

WHY IT MATTERS

Age alone is a bad signal — plenty of critical infrastructure has not been touched in years precisely because it works. Deleting on age produces an outage and ends the cleanup program. The safety check has to be part of the query, not a manual review of each candidate afterward.

WHAT LIVEASSIST RETURNED

LiveAssist returns objects untouched beyond the age threshold that also have no inbound references, no active leases, and no dependent records, with the dependency check shown so the reviewer can trust it. Four hundred objects qualify out of several thousand that are merely old. That distinction is the difference between a cleanup that ships and one that gets cancelled after the first incident.

#35 "Give me the estate summary for the ops review."

The recurring deliverable, produced without an engineer's afternoon.

weekly/monthly

Platform Ops

Access & Governance

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THE SITUATION

The monthly ops review needs the DDI section. Historically this costs an engineer most of an afternoon assembling numbers that will be read for ninety seconds.

WHY IT MATTERS

The manager needs consumption, growth, health, and exception counts in a form a non-DDI audience can follow. The engineer producing it is the most expensive person to assign to transcription, and because it is manual it gets skipped in busy months, which is when the numbers would have mattered most.

WHAT LIVEASSIST RETURNED

LiveAssist produces the summary directly: address utilization by region with month-over-month change, DNS record counts and growth, deploy success rate, open hygiene exceptions, and the notable movements with a sentence of context on each. The manager gets it before the meeting without an engineer touching it, and because it is cheap it actually gets produced every month, which is what makes the trend line worth anything.

#36 "Who has rights over this — and who granted them, when?"

Access attestation. The grant provenance is the half auditors actually want.

monthly

Access & Governance

Security

Hide the story ▲

THE SITUATION

Access attestation season. The auditor's question is not just who has rights over this zone today, but who granted them and when — and the second half is the part nobody can produce.

WHY IT MATTERS

Current-state permissions are visible. The provenance of those permissions — who assigned this role, on what date, under what approval — is what an attestation actually requires, and it lives in change history that has to be correlated to the access model by hand. The result is that attestations are signed on incomplete evidence.

WHAT LIVEASSIST RETURNED

LiveAssist returns the effective access on the object with the grant history attached to each entry: the principal, the role, where it was inherited from if it was not set directly, the date it was granted, and the account that granted it. Two entitlements trace back to a departed employee's account granting access outside the normal process. The attestation is defensible, and it surfaced something worth remediating rather than just producing a signature.

#37 "How is address consumption trending — do we need space this budget cycle?"

Procurement timing, with evidence rather than instinct.

quarterly

IPAM

Access & Governance

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THE SITUATION

The service owner has to decide whether to ask for address space in this budget cycle. Asking and not needing it damages credibility; needing it and not asking means an emergency procurement at a bad price.

WHY IT MATTERS

The trend is the deliverable and nothing produces it. Point-in-time utilization is available but the historical series is not retained in a queryable form, so the argument to finance is made on instinct and finance responds to instinct the way it always does.

WHAT LIVEASSIST RETURNED

LiveAssist returns consumption over time — allocated versus used, by region and business unit, with growth rates and a projection. Overall consumption is up 9% year over year, but two regions account for nearly all of it and one of them exhausts in eight months. The budget request is scoped to those two regions with a projection behind it, which is both cheaper and far more likely to be approved than a general ask.

#38 "Which business units consume the most address space and DNS records?"

Chargeback/showback, and the argument for cleanup funding.

quarterly

IPAM

Access & Governance

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THE SITUATION

The cleanup program needs funding and the service owner needs an argument. "DNS is messy" has never once moved a budget. "This business unit consumes 40% of the address space and 60% of the record churn" does.

WHY IT MATTERS

Showback requires attributing consumption to an owner, which requires the tag or UDF hierarchy to be joined to consumption metrics and rolled up. Nobody does this because it is a multi-hour data exercise that has to be repeated every quarter to stay current.

WHAT LIVEASSIST RETURNED

LiveAssist rolls consumption up the ownership hierarchy: address space allocated and actually used per business unit, DNS record counts, and change volume, with quarter-over-quarter movement. Two business units account for most of the growth and one of them has the worst hygiene exception rate in the estate. That pairing — heaviest consumer, worst hygiene — is the funding argument, and it took a query rather than a project.

#39 "Are we meeting our DDI SLAs — request-to-provision time, deploy success rate?"

The board slide, and the case for headcount.

monthly

Platform Ops

Access & Governance

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THE SITUATION

The service owner is asked to justify headcount. The only available evidence is that the team is busy, which is what every team says.

WHY IT MATTERS

DDI service levels are real and measurable — how long from request to provisioned address, what fraction of deploys succeed on the first attempt, how quickly hygiene exceptions are closed — but nothing measures them, so the team is managed on anecdote. Without the numbers, the case for headcount competes badly against teams that have dashboards.

WHAT LIVEASSIST RETURNED

LiveAssist derives the SLA metrics from timestamps already in the data: request-to-provision intervals from object creation and modification history, deploy success rate from deployment records, and exception aging from the hygiene sweeps. Median provision time has doubled in two quarters while volume rose 40%. That is a headcount case with a trend line, and it is derived rather than self-reported, which is why it survives scrutiny.

#40

"Which DNS records point at IPs we no longer control?"

Subdomain-takeover exposure. A pure read that maps directly to a real attack.

monthly

Security

DNS

Hide the story ▲

THE SITUATION

The security team is working through subdomain-takeover exposure after a peer company was hit by exactly this. The question is which DNS records point at addresses the organization no longer controls.

WHY IT MATTERS

Cloud resources get released and DNS records do not get cleaned up. The dangling record points at an IP that anyone can now claim, and claiming it means serving content from a trusted name. It maps to a real, cheap, frequently exploited attack, and finding it requires joining DNS records to current address ownership — two systems that do not talk.

WHAT LIVEASSIST RETURNED

LiveAssist joins the DNS record inventory against the address space the organization actually holds, including current cloud allocations, and returns every record whose target falls outside it. Nine records point at IPs in ranges released back to a cloud provider months ago, three of them under names that still carry the corporate brand. All nine are removed the same week, closing a live takeover path that no scanner had flagged.

#41 "Are internal and external views consistent where they should be, isolated where they must be?"

Split-horizon leaks expose internal topology; nobody checks systematically.

quarterly

Security

DNS

[Hide the story ▲](#)

THE SITUATION

An internal architecture review raises a question nobody has checked: are the internal and external DNS views actually isolated where they are supposed to be, and consistent where they are supposed to be?

WHY IT MATTERS

Split-horizon is a security control, and like every control it degrades. A record added to the wrong view leaks internal hostnames and topology to anyone who asks. A record missing from a view that should have it breaks a service. Both drift from ordinary work, and nothing verifies the intended posture because the intent has never been expressed in a checkable form.

WHAT LIVEASSIST RETURNED

The policy is expressed as the query — these zones must be isolated, these must be consistent — and LiveAssist returns the violations rather than a dump to be reviewed. Six internal-only names are present in the external view, including two that reveal internal subnet structure in their naming. The review turns from a discussion into a remediation list, and the same query becomes a recurring check.

#42 "Which zones are DNSSEC-signed, and are any signatures near expiry?"

An expired signature is a self-inflicted outage with a countdown on it.

monthly

Security

DNS

[Hide the story ▲](#)

THE SITUATION

A peer organization had a multi-hour outage when a DNSSEC signature expired over a holiday weekend. The security analyst wants to be certain that cannot happen here.

WHY IT MATTERS

An expired signature is a self-inflicted outage with a visible countdown on it, and the countdown is only visible if someone looks. Signing status and expiry live per-zone, so verifying the whole estate means opening every zone, which means it is verified once after an audit finding and never again.

WHAT LIVEASSIST RETURNED

LiveAssist returns signing posture across every zone: which are signed, which are not and whether they should be, key rollover state, and signature expiry dates sorted by proximity. Two zones have signatures expiring within eighteen days and one has a KSK rollover that stalled halfway through. All three are remediated on a Tuesday instead of over a holiday weekend, and the query goes into the scheduled pass so the countdown is never unobserved again.

#43 "Evidence pack: who could change DNS in scope X during the audit period, and what did they change?"

SOC2/PCI evidence currently assembled by hand over weeks.

quarterly

Security

Access & Governance

[Hide the story ▲](#)

THE SITUATION

SOC 2 fieldwork. The auditor wants, for a defined scope and period, who had the ability to change DNS and what they actually changed. Last year this took three weeks of an engineer's time and the evidence was still challenged.

WHY IT MATTERS

The evidence has two halves and they live in different places: the access model shows who could, the change log shows what happened, and correlating them across an audit period is manual joining work. Because it is expensive it is done once a year, badly, under deadline pressure.

WHAT LIVEASSIST RETURNED

LiveAssist returns both halves joined and scoped: every principal with change rights over the scope during the period including access that was granted and revoked mid-period, and every change each of them made, with timestamps and before/after values. What took three weeks becomes a query, and the evidence pack is stronger because it captures the mid-period access changes the manual process always missed.

#44 "Which address space is in PCI/CDE scope, and has anything entered or left it?"

Scope creep is an audit finding; detecting it is a query, not a project.

quarterly

Security

IPAM

[Hide the story ▲](#)

THE SITUATION

PCI scope is defined by a set of address ranges. Between assessments, things move. The analyst needs to know whether anything has entered or left the cardholder data environment since the last review.

WHY IT MATTERS

Scope creep is a finding. A workload that quietly acquires an address inside the CDE range brings itself into scope with all the controls that implies, and nobody notices until an assessor

does. Detecting it means comparing the current inhabitants of a set of ranges against the last known state — a query, but one nobody has set up.

WHAT LIVEASSIST RETURNED

The CDE ranges are expressed as the scope of the question and LiveAssist returns what is in them now, what was in them at the last assessment date, and the delta with the responsible owner for each change. Four new hosts entered CDE ranges, three of them from a test environment that was never meant to be there. Scope creep caught between assessments, when it is a fix rather than a finding.

#45 "Every record created outside the change window or without a linked ticket."

Unauthorized-change detection. Needs BAM joined to the ticket system.

monthly

Security

Access & Governance

[Hide the story ▲](#)

THE SITUATION

The change policy requires every DNS change to reference a ticket and land inside a change window. Compliance with that policy has never been measured, only asserted.

WHY IT MATTERS

Unauthorized change is a control failure that is invisible unless someone systematically checks, and checking means joining BAM change history to the ticketing system and to the change calendar. Three systems, no join, so the control is monitored by trust.

WHAT LIVEASSIST RETURNED

LiveAssist joins the change log to the ticket system and the change calendar and returns the exceptions: changes with no referenced ticket, changes referencing a closed or unrelated ticket, and changes made outside an approved window. Thirty-one exceptions in the quarter, twenty-eight from one automation account that was never onboarded to the ticket integration. That is a process fix rather than a people problem, and it would never have been visible without the join.

#46 "Which service accounts and integrations hold write access, and when did each last use it?"

Least-privilege review; dormant write credentials are the standing risk.

quarterly

Security

Access & Governance

[Hide the story ▲](#)

THE SITUATION

Least-privilege review. The analyst is less interested in human accounts — those get reviewed at offboarding — than in the service accounts and integrations that hold write access and never leave.

WHY IT MATTERS

Dormant write credentials are the standing risk in every estate. An integration built for a project that ended two years ago still holds the rights it was granted, and nobody revokes it because nobody is certain it is unused. Certainty requires last-use data alongside the entitlement, and the two are not presented together.

WHAT LIVEASSIST RETURNED

LiveAssist reads Integrity's access model and returns every non-human principal with write access, the scope of that access, and the timestamp of its last actual write. Eleven service accounts hold write rights; four have not written anything in over eighteen months, and one of those was provisioned for a vendor engagement that concluded in 2023. Four credentials revoked with evidence rather than argument, and the standing risk drops by a third.

P8 — Cloud & automation engineer

· BAM is one source of truth among several now; reconciling is a daily read problem.

#47 "Does BAM match what Terraform state and our cloud IPAM believe?"

Multi-source-of-truth drift. The defining DDI problem of the cloud era.

daily

Automation & Cloud

IPAM

[Hide the story ▲](#)

THE SITUATION

A Terraform apply fails because the address it wanted is already allocated. Not in Terraform state — in BAM, by someone who used the UI. This is the third time this month.

WHY IT MATTERS

BAM, cloud provider IPAM, and Terraform state each believe they are authoritative and each is partly right. Drift between them is the defining DDI problem of the cloud era, and it surfaces as failed pipelines and duplicate allocations rather than as a drift alert, because nothing compares the three.

WHAT LIVEASSIST RETURNED

LiveAssist reads BAM and reconciles it against Terraform state and cloud IPAM, returning the three-way delta: allocations present in BAM but absent from state, resources live in the cloud with no BAM record, and addresses where the three disagree on ownership. Forty-seven discrepancies, clustered in two accounts that were onboarded before the pipeline existed. The reconciliation runs daily thereafter and the failed applies stop.

#48 "Which subnets are automation-managed vs. hand-edited?"

Tells you what's safe to reconcile and what a pipeline will fight over.

weekly

Automation & Cloud

IPAM

[Hide the story ▲](#)

THE SITUATION

Before pointing a reconciliation pipeline at a set of subnets, the engineer needs to know which of them are automation-managed and which are hand-curated by a team that will not appreciate a robot rewriting their work.

WHY IT MATTERS

Running automation against a hand-managed subnet produces a fight: the pipeline reverts a human's change, the human reverts it back, and trust in the automation dies. The distinction between managed and unmanaged exists in practice but not as data, so it is maintained in a wiki page that is always slightly wrong.

WHAT LIVEASSIST RETURNED

LiveAssist infers and reports management posture from provenance: which objects were created and last modified by automation accounts versus named users, how mixed each subnet is, and where the boundary between the two currently sits. Six subnets thought to be automation-managed have recent human edits, meaning a reconciliation run would have reverted live changes. The pipeline scope is corrected before the first run rather than after the first incident.

#49 "Before my pipeline allocates: is this block free, and am I entitled to it?"

A precondition check that converts a failed apply into a clean skip.

per-run

Automation & Cloud

IPAM

[Hide the story ▲](#)

THE SITUATION

A pipeline is about to allocate a block. Before it does, it needs two answers: is the block actually free, and does this pipeline's credential have the right to take it?

WHY IT MATTERS

Without the precondition check, the pipeline discovers the answer by failing mid-apply, which leaves partial state and requires a human to unwind it. A clean skip is cheap; a failed apply is not. And entitlement is the half that is usually skipped, so the failure arrives late and looks like a permissions bug rather than a design gap.

WHAT LIVEASSIST RETURNED

LiveAssist answers both in one precondition read: the block's current allocation state including reservations and pending changes, and what the calling credential is actually permitted to do with it. The block is free but the credential is scoped to a different parent. The pipeline skips cleanly, logs a specific reason, and a human fixes the scoping — instead of a half-applied change and an hour of cleanup.

#50 "What's the machine-readable shape of what I'm about to read?"

Lets a pipeline bind to the data without hardcoding today's field names.

per-run

Automation & Cloud

Agent & AI

[Hide the story ▲](#)

THE SITUATION

A pipeline is being written against LiveAssist. The developer needs to know the shape of the data — field names, types, which fields are optional — without hardcoding whatever today's response happens to contain.

WHY IT MATTERS

Hardcoding against an observed response works until a field is renamed or an optional field is absent, and then the pipeline breaks in production for reasons that look like a data problem. Every integration built without a schema carries this fragility, and the cost lands on whoever is on call rather than whoever wrote it.

WHAT LIVEASSIST RETURNED

LiveAssist describes the underlying Integrity object model: the field names, field names and types, cardinality, and which fields are guaranteed versus conditional. The pipeline binds to the described schema and validates against it, so a change in the underlying model produces a clear validation error at the boundary rather than a silent misparse three steps downstream.

#51 "Which records were created by automation and are now orphaned from their stack?"

Cloud garbage collection; ephemeral infra leaves permanent DNS behind.

weekly

Automation & Cloud

DNS

[Hide the story ▲](#)

THE SITUATION

An ephemeral test environment was destroyed last week. Its DNS records are still there. So are the records from the fourteen environments before it.

WHY IT MATTERS

Cloud infrastructure is disposable and DNS is not. Every stack that creates records and is then torn down leaves them behind unless the teardown is perfect, and teardown is never perfect. The accumulation is invisible because each individual leftover looks like a legitimate record, and the only way to tell is to check whether the stack that created it still exists.

WHAT LIVEASSIST RETURNED

LiveAssist returns records created by automation with their originating stack or workload identifier, joined against which of those stacks still exist. Two hundred and thirty records reference stacks that were destroyed, some over a year ago, and eleven of them point at addresses that have since been reassigned to other workloads — which is both garbage and a misdirection risk. Garbage collection with the evidence to justify each deletion.

#52 "Notify me when anything in this block or zone changes."

Event-driven automation instead of polling on a timer.

continuous

Automation & Cloud

[Hide the story ▲](#)

THE SITUATION

An automation workflow needs to react when address space in a particular block is allocated. Today it polls every five minutes, which means it is either late or wasteful, and usually both.

WHY IT MATTERS

Polling is the tax paid for the absence of events. It adds latency proportional to the interval, load proportional to the inverse, and complexity in the form of state tracking to detect what changed since last time. Every integration reimplements the same change-detection logic slightly differently.

WHAT LIVEASSIST RETURNED

LiveAssist supports subscription over the Integrity MCP server: the workflow registers interest in a block or a zone and receives a notification when anything within it changes, with the changed object and the nature of the change. Reaction time drops from minutes to seconds, the polling load disappears, and the same mechanism upgrades the scheduled health pass and the daily drift check from batch jobs to real-time.

P9 — M&A / integration architect · Low frequency, extreme stakes — every acquisition opens the same two questions.

#53 "Where does our address space collide with theirs?"

The opening question of every acquisition and every partner VPN.

per-project

M&A / Integration

IPAM

[Hide the story ▲](#)

THE SITUATION

An acquisition closes in ninety days. The integration architect has the acquired company's address documentation — a spreadsheet — and needs to know where it collides with the existing estate.

WHY IT MATTERS

This is the opening question of every acquisition and every partner VPN, and it gates everything downstream: the routing design, the firewall policy, the migration sequence, and the budget. Getting it wrong means discovering the collision during cutover, which is when it costs the most and is most visible.

WHAT LIVEASSIST RETURNED

LiveAssist compares the acquired ranges against the full existing hierarchy and returns the overlaps with both sides identified, whether each existing range is actually in use or merely allocated, and how many hosts sit in the contested space. Eleven overlaps, but only four involve ranges that are live on both sides. The integration plan is scoped against four real problems instead of eleven theoretical ones, in an afternoon rather than a month.

#54 "What's the smallest re-IP that resolves the collisions — and what breaks?"

Turns a six-month integration guess into a costed plan.

per-project

🔥 frontier

M&A / Integration

IPAM

[Hide the story ▲](#)

THE SITUATION

The overlaps are known. Now the question is the expensive one: what is the smallest renumbering that resolves them, and what breaks when it happens?

WHY IT MATTERS

Integration timelines are built on a guess at this number, and the guess is usually wrong by a factor that shows up as a schedule slip six months in. Answering it properly means simulating the renumber and tracing the dependency graph of everything affected — DNS records, DHCP reservations, firewall objects, hardcoded references — which nobody can do by hand at scale.

WHAT LIVEASSIST RETURNED

LiveAssist simulates candidate renumbering plans as a read: for each option, which ranges move, how many hosts are affected, and the dependency-ordered set of what breaks — DNS records requiring updates, reservations to rebuild, and references that will dangle. Moving two of their ranges instead of one of ours affects 340 hosts instead of 2,100. A six-month guess becomes a costed plan with an evidence trail, and the change board approves it in one sitting.

#55

"Which of their DNS names conflict with ours, and which are actually in use?"

Namespace merge planning; "in use" is what separates the hard cases from the noise.

per-project

M&A / Integration

DNS

[Hide the story ▲](#)

THE SITUATION

Both organizations use similar naming conventions and both have an intranet, a wiki, and a dozen service names in common. The namespace merge needs a conflict list, and more importantly a list of which conflicts actually matter.

WHY IT MATTERS

A raw conflict list is mostly noise — plenty of duplicated names on both sides are dead records nobody has queried in years. Treating all conflicts as equal turns a two-week problem into a six-month one, and the discriminator is whether the name is genuinely in use, which is not visible from the record itself.

WHAT LIVEASSIST RETURNED

LiveAssist returns the name conflicts across both namespaces annotated with evidence of use on each side: whether the target resolves, whether the address has a live lease or recent activity, and query volume where telemetry is available. Two hundred conflicts reduce to nineteen that are live on both sides and need a real decision. The other 181 are cleanup, and separating them is the difference between a merge that ships on time and one that does not.

#56

"What can I read here, and what does each thing mean?"

Semantic discovery so the agent doesn't guess at the object model.

per session

Agent & AI

Hide the story ▲

THE SITUATION

An agent connects to the estate for the first time and needs to understand what it is looking at before it can plan anything — not just which calls exist, but what the returned objects mean and how they relate.

WHY IT MATTERS

An agent without semantic grounding guesses at the object model, and its guesses are plausible and wrong in ways that are hard to detect. It will conflate a network with a block, assume a view is a zone, or treat an allocated address as a live one. Every such error propagates into a plan that looks reasonable and is not.

WHAT LIVEASSIST RETURNED

LiveAssist describes the Integrity model semantically: the object types, what each represents in DDI terms, how they nest and reference each other, and which fields carry meaning versus which are administrative. The agent builds a correct model of the estate before its first substantive read, and its subsequent questions are well-formed rather than exploratory — which is the difference between an agent that is useful and one that is merely confident.

#57

"How fresh is this answer, and where did it come from — live, mirror, or cache?"

Determines whether the agent may act on it or must re-read live.

per call

Agent & AI

Hide the story ▲

THE SITUATION

An agent is deciding whether it can act on an answer it just received. The answer looks fine. But if it came from a mirror that last synced three days ago, acting on it could be actively harmful.

WHY IT MATTERS

A stale capacity number, a stale lease, or a stale record state is worse than no answer, because the agent will proceed with confidence. Freshness is not a nicety in an agentic context; it is the precondition for the safety of everything downstream. And the agent cannot infer it — it has to be told.

WHAT LIVEASSIST RETURNED

Every LiveAssist response carries provenance: whether it came from a live read, a local mirror, or a cache, the age of the underlying data, and a confidence indication. The agent sees that the utilization figure is from a mirror synced six hours ago — fine for a report, not fine for an allocation — and re-reads live before acting. The metadata is cheap to attach and it is what makes every other answer on this page trustworthy.

#58

"What am I entitled to see and do with this credential?"

Stops the agent planning work it will be refused halfway through.

per session

Agent & AI

Security

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THE SITUATION

An agent is planning a multi-step task across several zones and subnets. Halfway through execution it will discover that its credential cannot see one of them — unless it asks first.

WHY IT MATTERS

An agent that plans work it will be refused partway through leaves partial state, produces confusing errors, and burns the user's trust. It also plans dishonestly, presenting a plan it has no ability to complete. The entitlement information exists; it is simply not exposed as something the caller can ask about itself.

WHAT LIVEASSIST RETURNED

LiveAssist answers what this credential may see and do, scoped to the objects in question, before the plan is formed. The agent learns it can read all four zones but has no visibility into one subnet, adjusts the plan to exclude it, and tells the user up front which part of the task needs a different credential. The plan that gets presented is the plan that can actually be executed.

#59 "If I made this change, what would it touch and what would resist?"

The read that must precede every write; the whole safety story depends on it. The retire-a-host slice ships today as the dry-run decommission composite — proof the class is buildable.

per task

frontier

Agent & AI

DNS

IPAM

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THE SITUATION

The agent has been asked to retire a host. It has assembled a change set. Before it proposes anything to a human, it needs to know what that change would touch and what would resist it.

WHY IT MATTERS

This is the read that must precede every write, and the entire safety story of an agentic DDI workflow rests on it. Without it, the agent's proposal is a guess with good grammar, and the human reviewing it has no more information than the agent did. With it, the review is over evidence.

WHAT LIVEASSIST RETURNED

LiveAssist simulates the change against Integrity: the objects that would be modified or removed, the dependents that reference them, the deployments that would be triggered, and the conditions that would cause the change to fail — locked objects, insufficient rights, records the credential cannot see. The retire-a-host slice of this already ships as Integrity's dry-run decommission composite, which is proof this class of capability is buildable rather than aspirational. The agent presents an impact-annotated proposal and the human approves something they can actually evaluate.

#60 "Diff what I proposed against what the estate actually looks like now."

Post-write verification and idempotency — did it land, and only it?

per task

Agent & AI

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THE SITUATION

A change was applied. The agent now needs to verify that what landed is exactly what was proposed — no more, no less — before it reports success or moves to the next step.

WHY IT MATTERS

"The API returned 200" is not verification. Partial application, side effects from triggered deployments, and concurrent changes by other actors all produce an estate that differs from the proposal in ways the write response will not reveal. Without post-write verification an agent cannot be idempotent, and without idempotency it cannot safely retry.

WHAT LIVEASSIST RETURNED

The agent asks LiveAssist to diff its proposed change set against the estate as it now stands, and receives the delta: what landed as intended, what did not land, and anything that changed which was not in the proposal. Two records applied, one was blocked by a lock, and a third object changed because a deployment cascaded. The agent reports

precisely that, retries only the blocked item, and never re-applies work that already succeeded.

► **Nine read primitives the 60 imply**

► **If you only ask for three, and two observations worth carrying**

Source: internal brainstorm document, dated 2026-08-24. Throughout, "LiveAssist" is the interface the practitioner actually talks to; it reaches BlueCat Integrity through the Integrity MCP server, so no one has to build or maintain their own connection. Topic tags are an added lens on top of the persona structure — several asks span more than one topic (e.g. reclaiming address space touches IPAM, DNS, and DHCP at once).